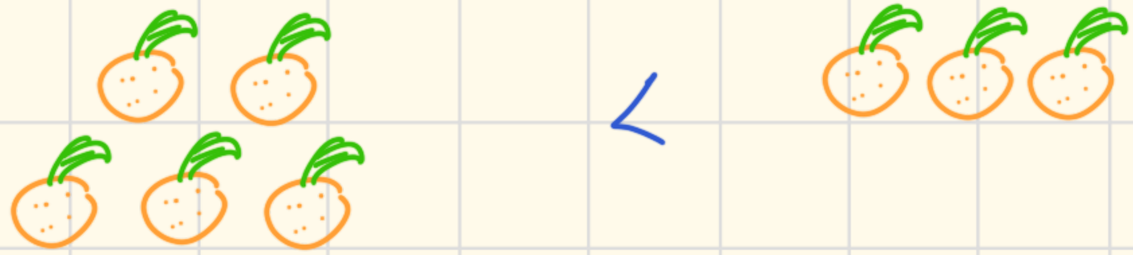
INEQUALITIES

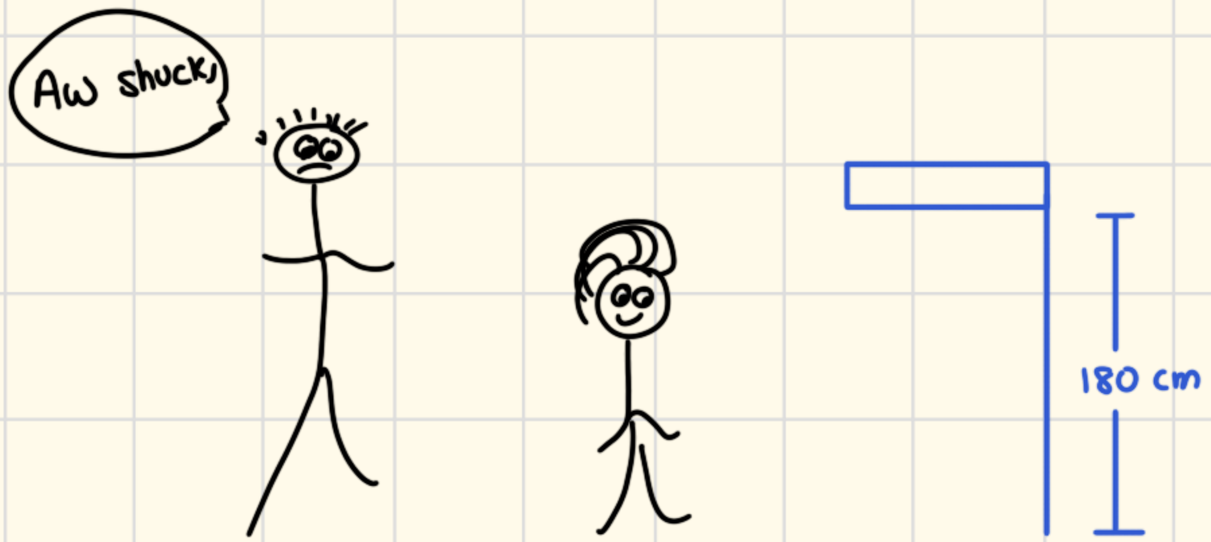
Inequalities represent 2 ideas.

1 Order



5 oranges **more** than 3 oranges

2 Condition



height > 180 < 180

limit to ride rollercoaster

INEQUALITIES

Symbols of inequalities and examples

$>$	greater than	$<$	less than
$\geqslant$	greater than or equal to	$\leqslant$	less than or equal to

TRUE STATEMENTS ✓	FALSE STATEMENTS ✗
$5 > 3$	$3 > 5$
$3 < 5$	$5 < 3$
$5 \geqslant 3$	$3 \geqslant 5$
$3 \leqslant 5$	$5 \leqslant 3$
$3 \geqslant 3$	$3 > 3$
$3 \leqslant 3$	$3 < 3$

INEQUALITIES

We will learn to find solutions to inequalities like :

$x < 5$

$x \geqslant 2$

NUMBER LINE

$3 < x < 5$

$x > 2$ and $x < -2$

COMPOUND

$x^2 < 4$

$x^2 - 5x - 6 < 0$

QUADRATIC

$y > x + 5$

$x + y < 4$

STRAIGHT - LINE (2D)

Inequalities

Inequalities are solved exactly like equation

but with an extra rule!

! Dividing OR Multiplying by a negative Switches signs !

Example 8

$$-3x + 2 < 0$$

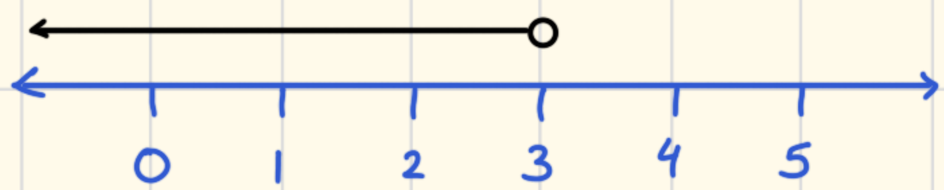
$$-3x < 2$$

$$x > \frac{2}{3}$$

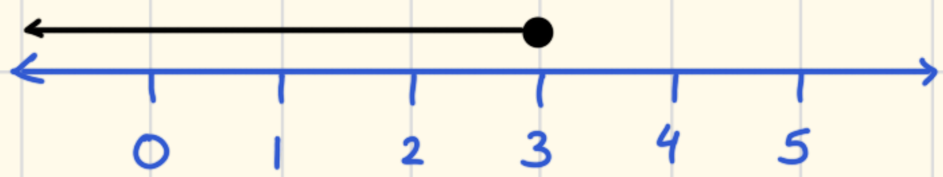
Number Line

We can draw solutions to inequalities on number lines!

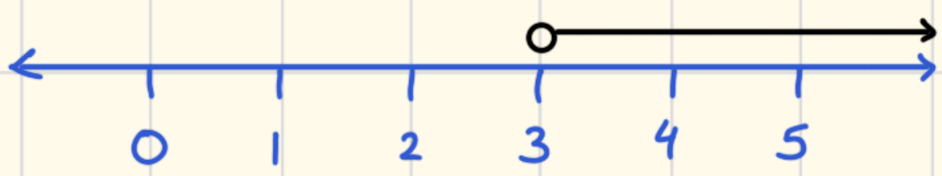
$x < 3$



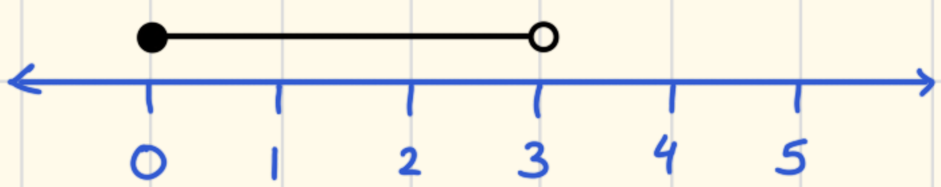
$x \leq 3$



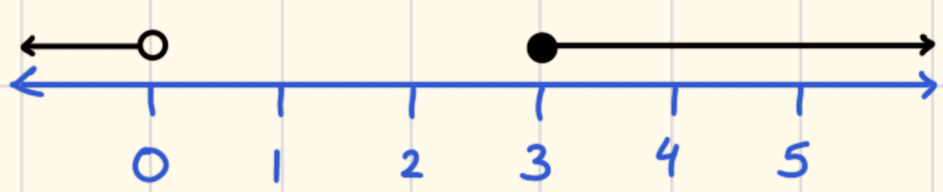
$x > 3$



$0 \leq x < 3$



$x < 0, \ x \geq 3$



Compound Inequalities

Compound inequalities are a combination of inequalities.

$$0 < 2x - 5 \leq 9$$

$+5 \qquad +5 \qquad +5$

$$5 < 2x \leq 14$$

$\div 2 \qquad \div 2 \qquad \div 2$

$$\frac{5}{2} < x \leq 12$$

Solution: $\left\{ x : \frac{5}{2} < x \leq 12 \right\}$ ← Set notation!

QUADRATIC INEQUALITIES

Quadratic inequalities are solved similarly to their equations.

1. Take everything on one side
2. Factor
3. If < 0, then x is between solutions.

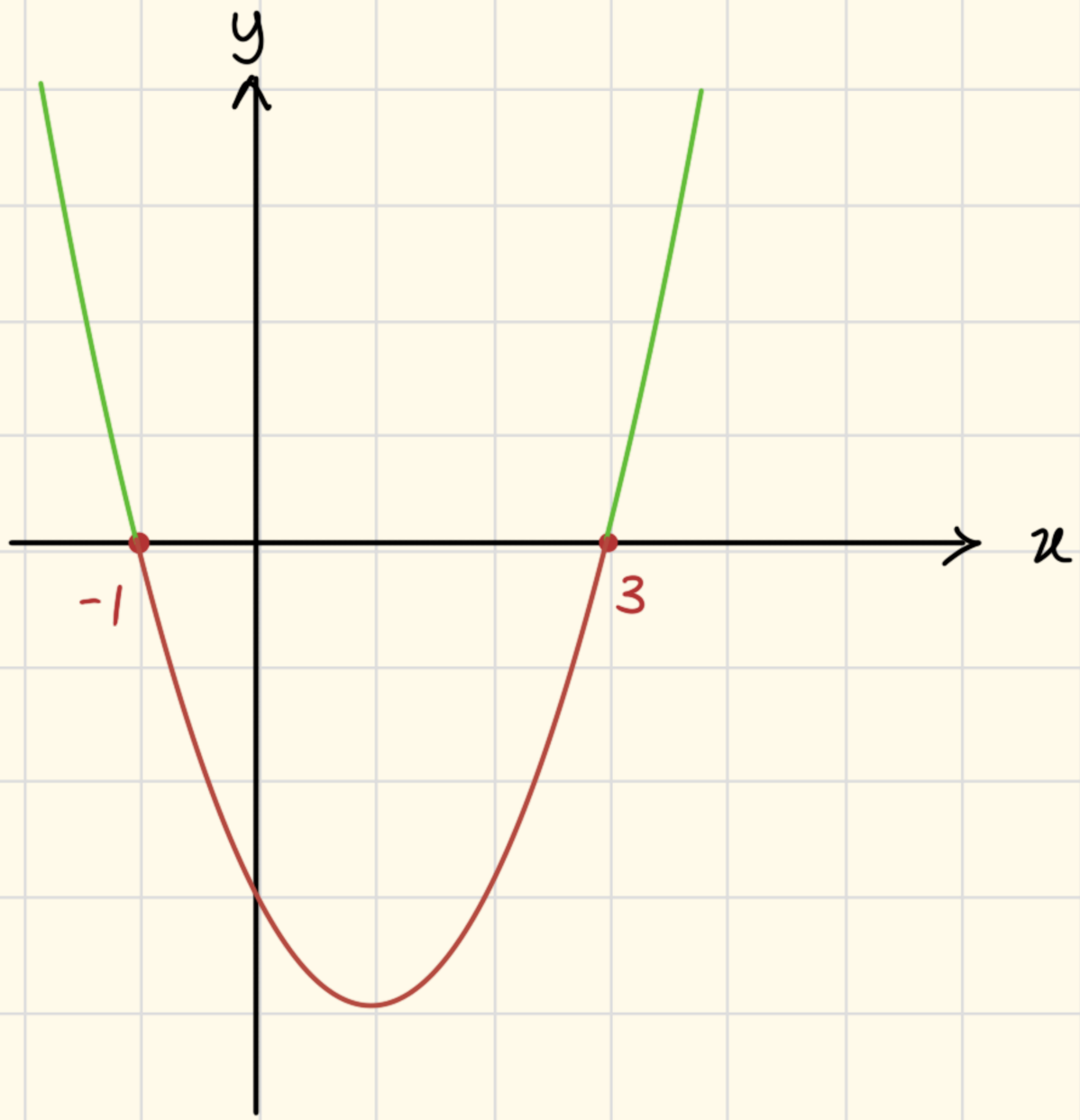
   > 0, then x is outside solutions.

1 $x^2 < 2x + 3$

$x^2 - 2x - 3 < 0$

2 $(x + 1)(x - 3) < 0$

3



> 0 $x<0,\ x>3$

< 0 $0 < x < 3$

STRAIGHT - LINE INEQUALITIES

Straight - line inequalities have solutions that are 2D (plane) not (1D) number line.

1. Make y subject of inequality
2. Draw line $y =$
3. Pick any point not on line.
   Check if it satisfies inequality
4. Shade region that satisfies inequality.

STRAIGHT - LINE INEQUALITIES

Draw solution to $x < 3$

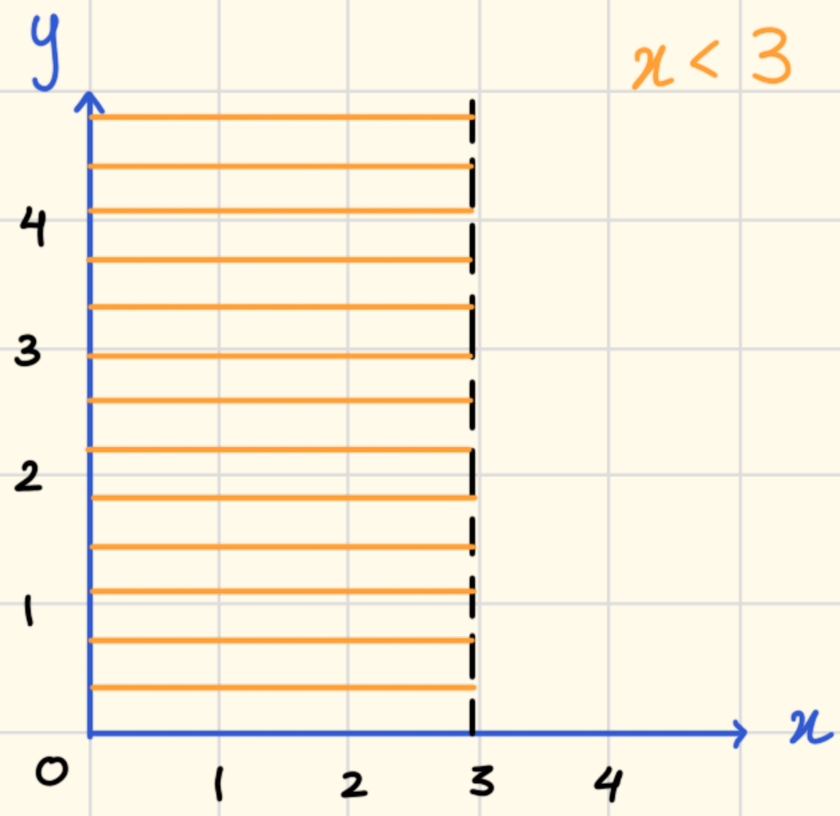
region left

since point

$(0, 0)$

$0 < 3$ ✓

satisfies



$x < 3$

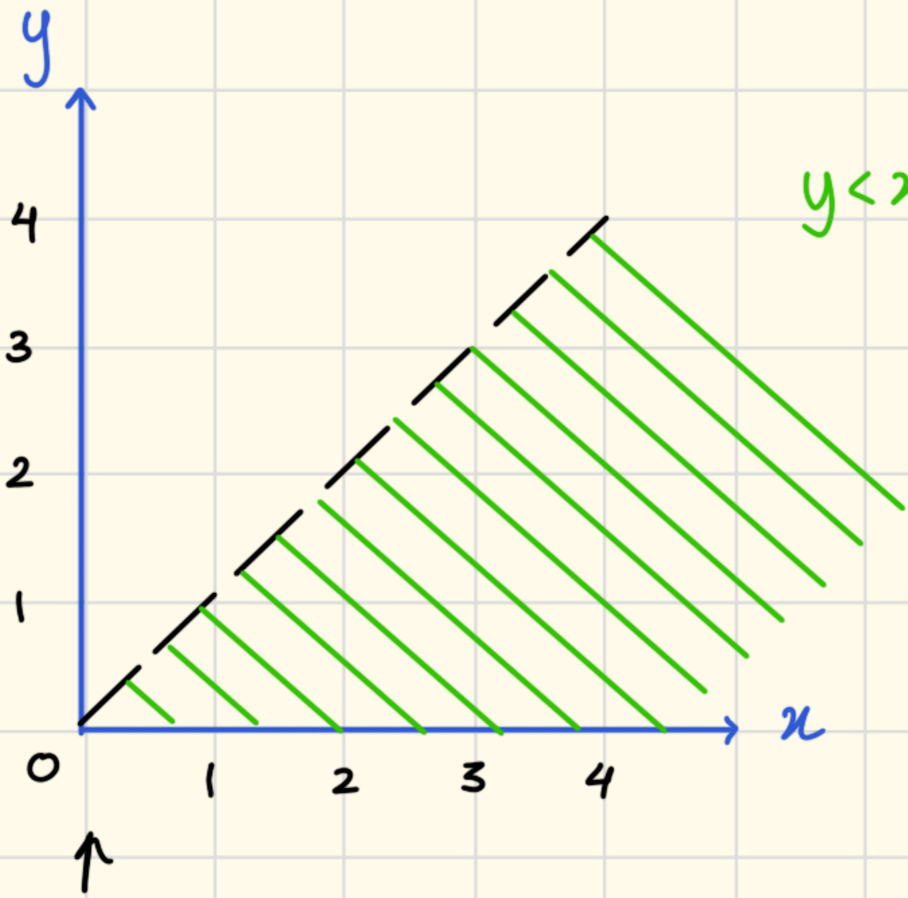
↑

dotted since

no equal

STRAIGHT - LINE INEQUALITIES

Draw solution to $y < x$



dotted since no equal

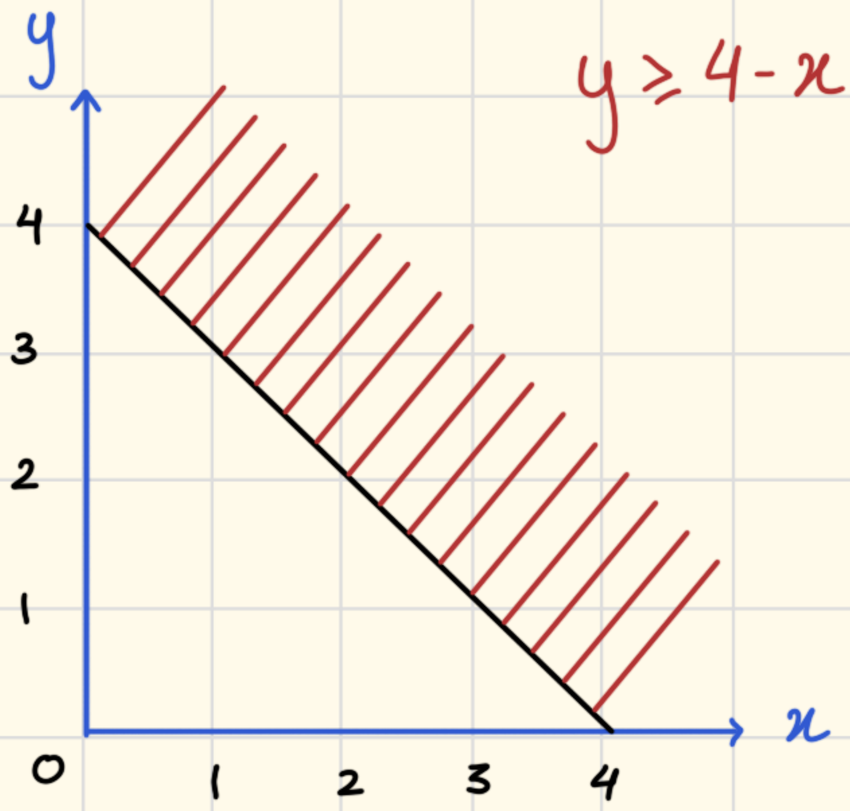
← region down since point (1, 0)

0 < 1 ✓

satisfies

Straight-Line Inequalities

Draw solution to $y \geq 4 - x$



$y \geq 4 - x$

← region up since point (0, 0)

$0 \geq 4$ ✗

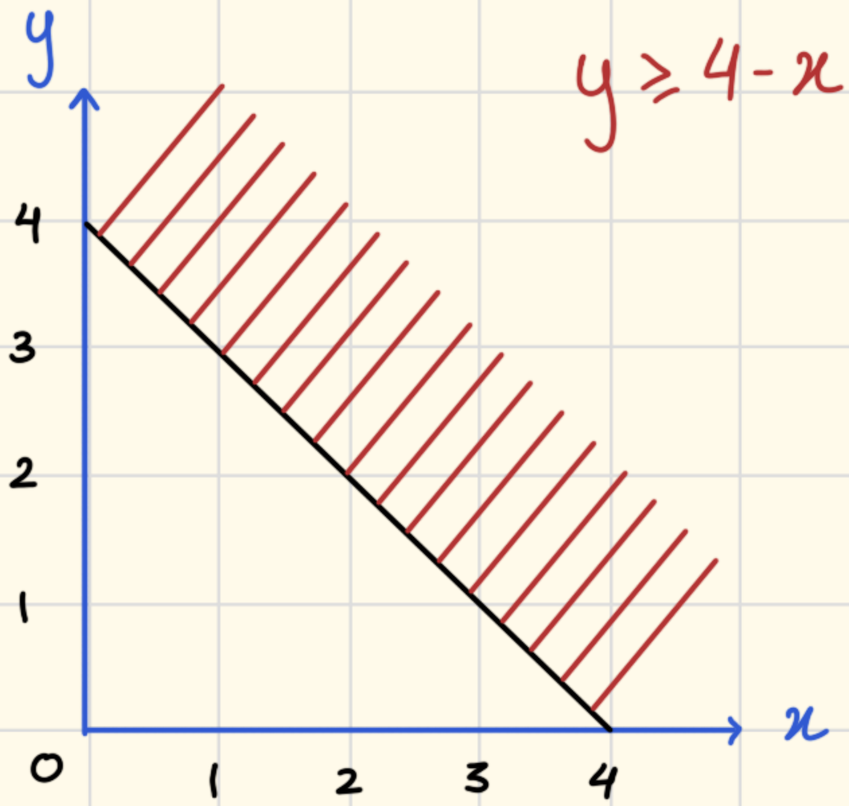
doesn't satisfies

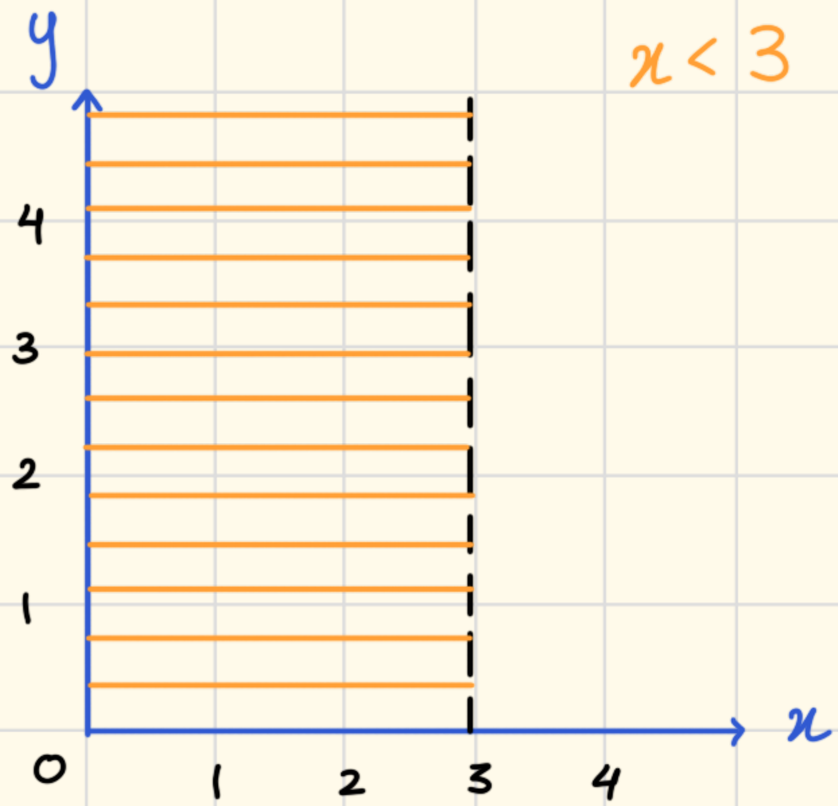
↑ solid since equal

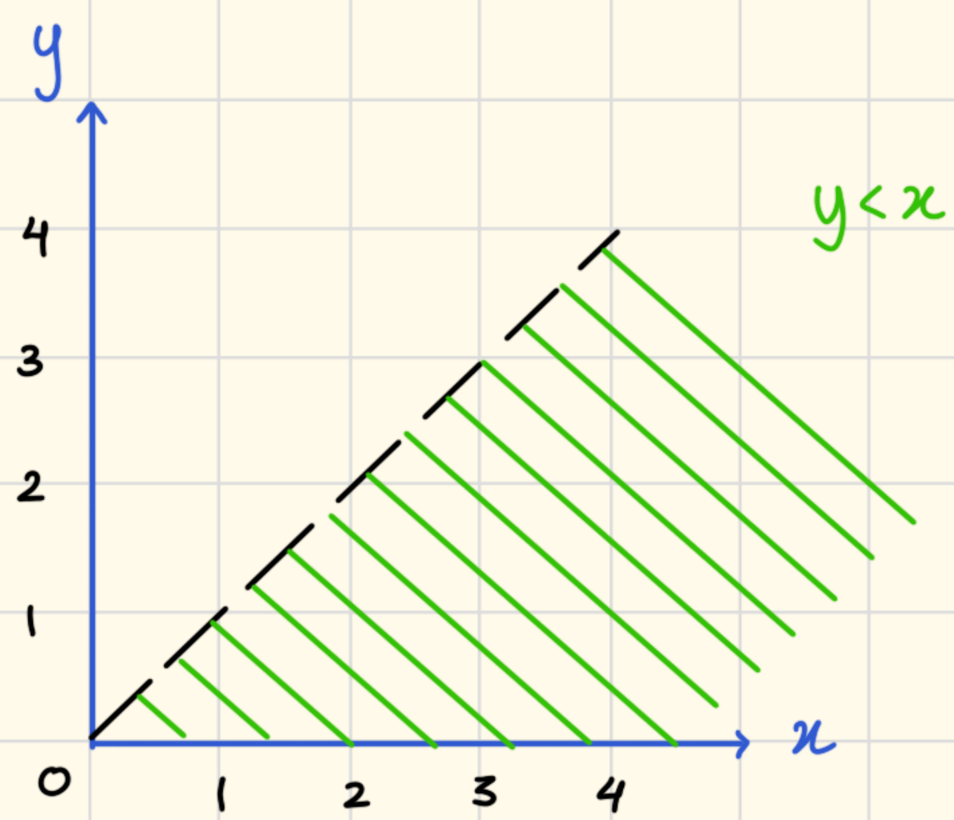
Straight-Line Inequalities Systems

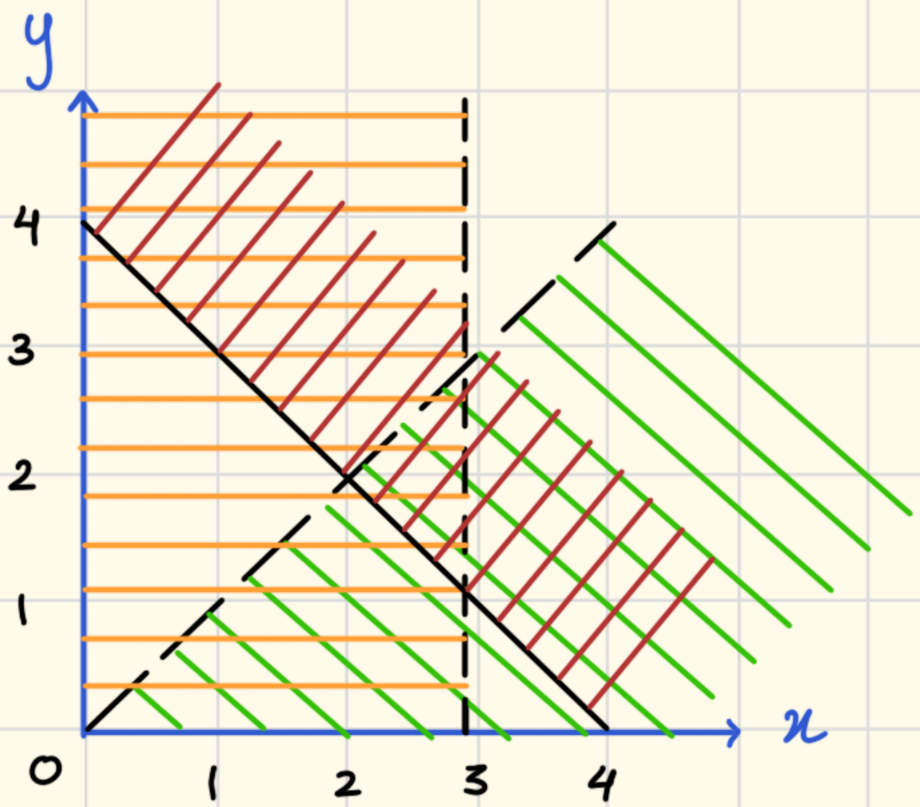
Systems of inequalities are solved by intersection of their solution regions

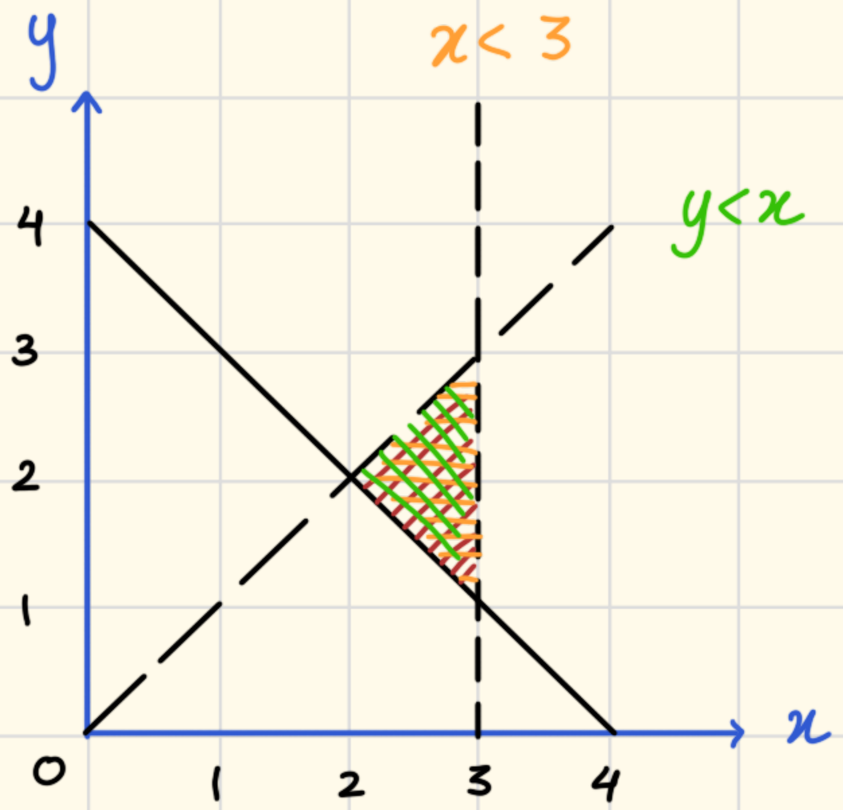
1. Draw lines
2. Check if dotted or solid
3. Shade region for each inequality
4. Shade ONLY the intersection of region.


y
x
0
1
2
3
4
y ≥ 4 - x


y
x
0
1
2
3
4
x < 3


y
x
0
1
2
3
4
y < x





$y \geq 4-x$

Does $(2.5, 2)$ satisfy the system?

y	$<$	3	$2 < 3$	✓
x	$<$	3	$2.5 < 3$	✓
y	$\geqslant$	$4-x$	$2 \geqslant 1.5$	✓